Faculty of Technology, University of Delhi

SEC - PROJECT - SEMESTER 7

Benchmarking Framework for LLM-Based Agent Interoperability

Measuring Interoperability Loss Across Heterogeneous Agent Boundaries

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Introduction

LLM Multi-Agent Systems

Multiple AI agents collaborate by exchanging tasks, context, and tool capabilities to solve complex problems.

Why Multi-Agent Systems?

- Complex tasks can be decomposed into specialized subtasks.
- Different agents can use different models, tools, and capabilities.
- Agents increasingly operate across vendors, frameworks, and protocols.
- This makes reliable agent-to-agent handoff essential

Key idea:

An agent can successfully communicate with another agent without successfully preserving the intent, context, constraints, and capabilities required to complete the task.

Bottom takeaway

Communication enables agents to interact. Interoperability determines whether they can work together correctly.

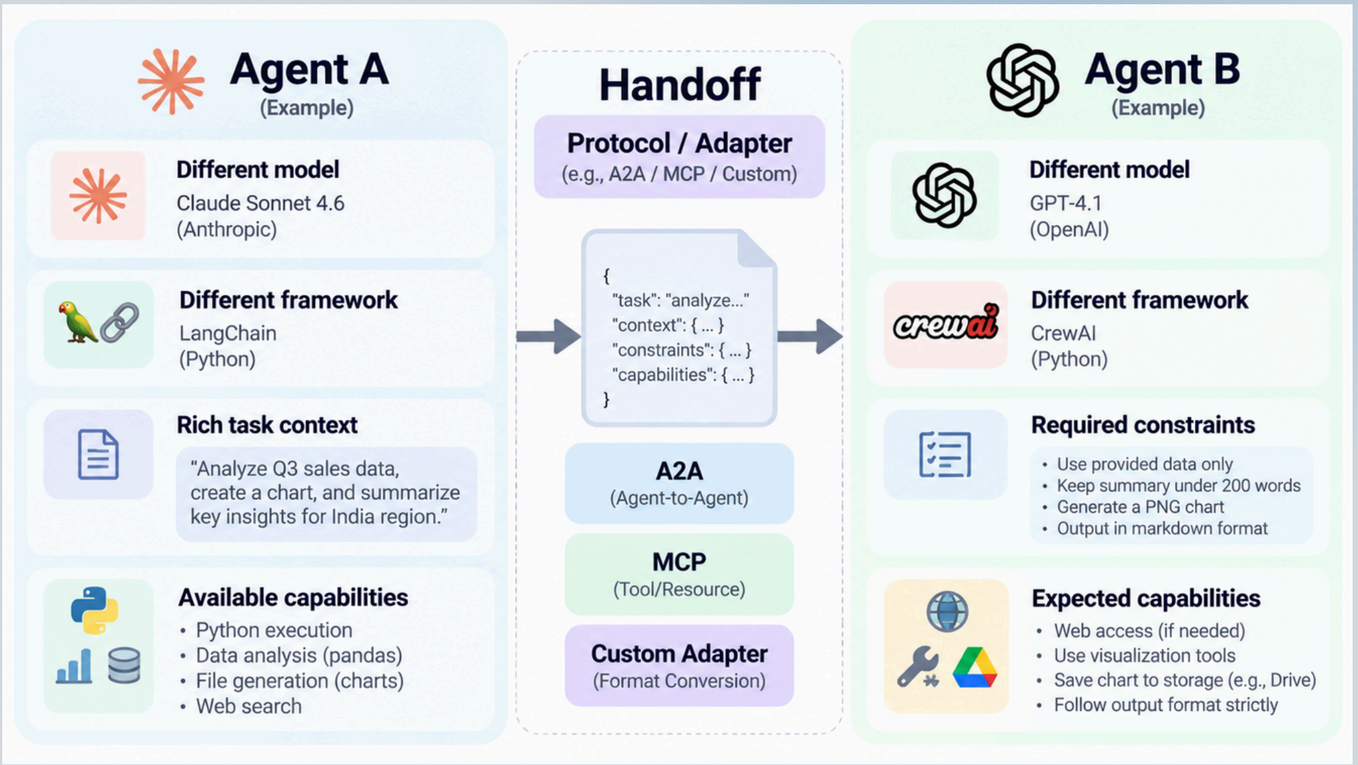


Fig 1 : The Interoperability Problem

What We Mean by Interoperability ?

MCP vs A2A

MCP

- Agent access to tools and resources
- Client-server architecture
- Standardizes tools, resources, and prompts
- Not designed for agent-to-agent coordination

A2A

- Agent-to-agent task delegation
- Agent Cards for capability discovery
- Supports task lifecycle and stateful interaction
- Complements MCP

Level	Meaning	Failure Example
Syntactic	Message format matches	Malformed JSON
Protocol	Handshake and lifecycle succeed	A2A state mismatch
Semantic	Intent and constraints preserved	"$500 limit" dropped
Operational	End-to-end performance preserved	Success rate drops

MCP solves tool-agent communication; A2A solves agent-agent delegation. Our benchmark operates at the A2A/handoff boundary, where task semantics must cross.

Literature Review

Work	What It Established	What It Missed
KQML (Finin et al., 1994)	Standardized agent communication; REQUEST, INFORM, SUBSCRIBE	No shared semantics or grounding
FIPA ACL (FIPA, 1997–2004)	Formal semantics; standardized communicative acts	Semantic grounding unresolved; not scalable to LLMs
AutoGen (Wu et al., 2023)	Programmable multi-agent conversations	Homogeneous framework; no interop measurement
MetaGPT (Hong et al., 2023)	Structured software development multi-agent systems	No protocol-level evaluation
CAMEL / ChatDev (2023)	Role-playing and task decomposition	Tightly coupled; no handoff analysis

📄 **18 Papers Reviewed Spanning 1994 – 2026**

Why This Matters ?

- Message delivery ≠ semantic agreement
- LLM frameworks assume agents interpret task specifications consistently.
- Existing work does not isolate the handoff boundary as a measurement target.
- The core semantic interoperability problem was not solved. It became implicit in LLM systems.

Agent/Multi-Agent Benchmarks and Failure Analysis

What Existing Benchmarks Measure and What They Don't

Benchmark	Focus	Key Limitation
AgentBench / AgentBoard	Single-agent task performance	No inter-agent handoff
MultiAgentBench	Multi-agent collaboration	No protocol/adapter comparison
MAST	Multi-agent failure taxonomy	No quantitative boundary metric
BFCL / ToolSandbox / τ-bench	Tool use and function calling	Single-agent
ToolBench / API-Bank	API discovery and planning	No cross-agent evaluation
MCP-AgentBench / MCP-Bench	MCP tool use	No agent-to-agent delegation

Existing benchmarks ask: Can an agent use tools?

Our benchmark asks: How much capability and task performance survive the handoff?

Key Findings

- AgentBench / AgentBoard: Measure single-agent capability.
- MultiAgentBench: Measures collaboration but does not vary the protocol or adapter.
- MAST: Identifies inter-agent misalignment but does not quantify boundary-level loss.
- MAST shows that ~79% of failures involve coordination/specification issues, but not how much performance is lost, where it occurs, or how protocols compare.

Timeline:
23 September– 13 December 2026

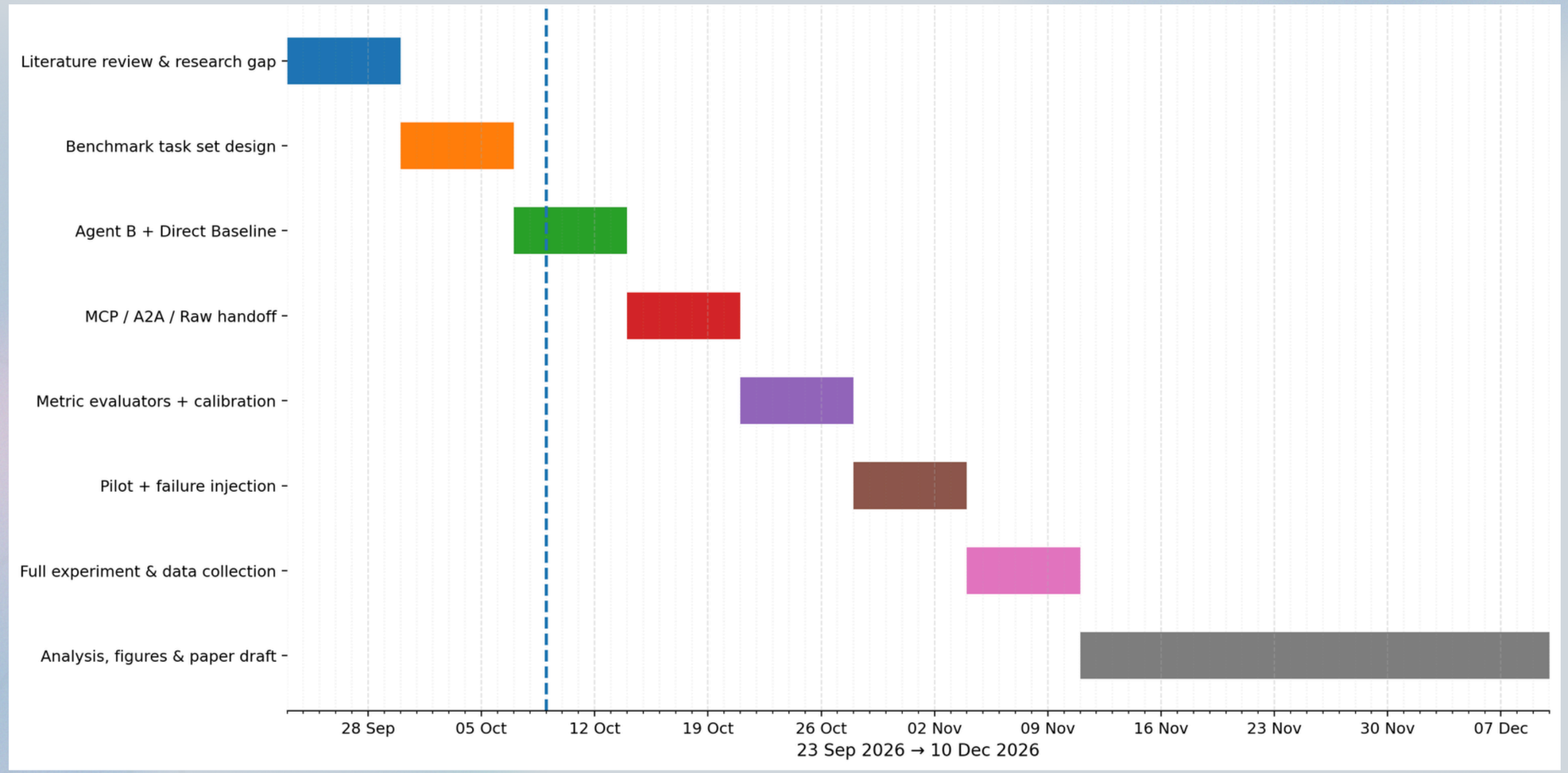


Fig 2: Project timeline graph

Conclusion & Summary

PROBLEM

Heterogeneous agents increasingly exchange tasks across protocols and vendors, but the performance cost of handoffs remains unmeasured.

RESEARCH GAP

No benchmark isolates the handoff boundary and measures its performance loss against a direct baseline.
Proposed Benchmark: InteropBench
Direct vs. Interoperated controlled comparison
6 metrics: TSR, CFS, CPR, CR, LO, TCO
Cross-vendor × cross-protocol: MCP, A2A, Raw
Reproducible task set with LLM-judge + human calibration

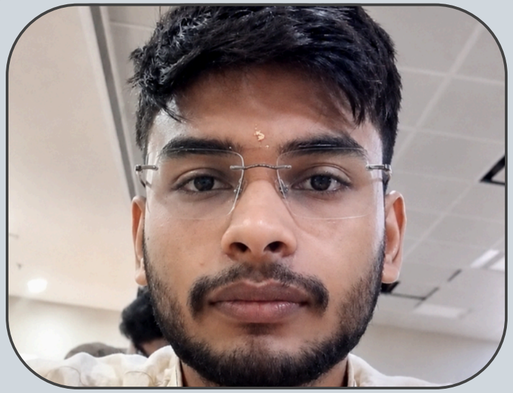
WHAT EXISTING WORK SHOWS

- MAST: Coordination is a major MAS failure source.
- ProtocolBench: Protocol choice affects performance.
- MCP-Bench / ToolBench: Tool use is well evaluated.
- Facts Without Rules / Model Switching: Handoffs can cause measurable performance loss.

EXPECTED CONTRIBUTION

- Measure performance loss caused by interoperability boundaries.
- Compare protocol × vendor configurations.
- Provide a reproducible benchmark for future protocols.
- Novelty: InteropBench measures Interoperability Loss (IL) by comparing direct and interoperated execution, with separate measures for context, constraints, capabilities, and tool fidelity.

Meet Our Team



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Thank You

for your time
and attention